



CSS Basics - Topics

- 1) HTML vs CSS vs JS
- 2) What is CSS?
- 3) History of CSS
- 4) Cascading Style Sheets
- 5) CSS Basic Syntax
- 6) Ways to apply css
- 7) Comments in CSS
- 8) Color Property
- 9) Background Color Property
- 10) Color Formats
- 11) Common Text Properties



HTML vs CSS vs JS

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What is **CSS**?

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History of CSS

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Cascading Style Sheets

CSS) is a stylesheet language used to describe the presentation of a document written in HTML



CSS **Basic** Syntax

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Syntax:

```
selector {  
  property: value;  
}
```

Example:

```
p {  
  color: blue;  
  font-size: 16px;  
}
```



Ways to **Apply** CSS

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There are three main ways to include CSS:

- Inline styles
- Using the `<style>` tag (internal stylesheet)
- External stylesheet



Inline Style

- Styles are written directly inside the HTML element using the style attribute.
- Best suited for quick testing or applying a single, unique style to one element.

Example: `<h1 style="color: red;">TheAtoms </h1>`



Using the **<style>** Tag

- Styles are grouped inside a single `<style>` element placed within the `<head>` section of the HTML document.
- Ideal for styling a whole page consistently without creating a separate CSS file.

Example

```
<style>  
  h1 {  
    color: red;  
  }  
</style>
```



Limitations of Inline Styles and `<style>` Tag

- Hard to Maintain
- Poor Scalability
- Limited Reusability
- Messy Codebase
- Breaks Separation of Concerns

We usually avoid using **inline styles** and the **`<style>`** tag in real-world projects.



External Style Sheet

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- Keeps HTML clean by separating structure and presentation.
- Makes it easy to reuse the same stylesheet across multiple pages for a consistent design.



How to **Link** a Stylesheet to HTML

Use the **<link>** tag inside the **<head>** section to include your stylesheet.

Syntax:

```
<link rel="stylesheet" href="style.css" />
```

rel: relationship with current document...

- stylesheet
- icon
- license



Comments in CSS

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Color Property

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- Defines the text (foreground) color of an element

Example:

```
h1 {  
  color: orange;  
}
```



Background Color Property

- Sets the background color of an element

Example:

```
div {  
  background-color: orange;  
}
```



Color Formats in CSS

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1. Named Colors
2. RGB
3. RGBA
4. HEX
5. HSL
6. HSLA



Named Colors

- CSS supports over 140 predefined color names like red, blue, and gold.
- Named colors are quick and easy to use, but offer less precision and flexibility than color codes like HEX or RGB.



RGB Color Model

- Represents colors by mixing Red, Green, and Blue light values, each ranging from 0 to 255
- Useful for defining millions of precise colors, e.g., `rgb(255, 0, 0)` for pure red

Example:

red is `rgb(255, 0, 0)`

green is `rgb(0, 255, 0)`

blue is `rgb(0, 0, 255)`



RGBA Color Model

- Extends RGB by adding an Alpha channel to control transparency
- (0 = fully transparent, 1 = fully opaque)

Example:

red is `rgb(255, 0, 0, 0.5)`

green is `rgb(0, 255, 0, 0.4)`

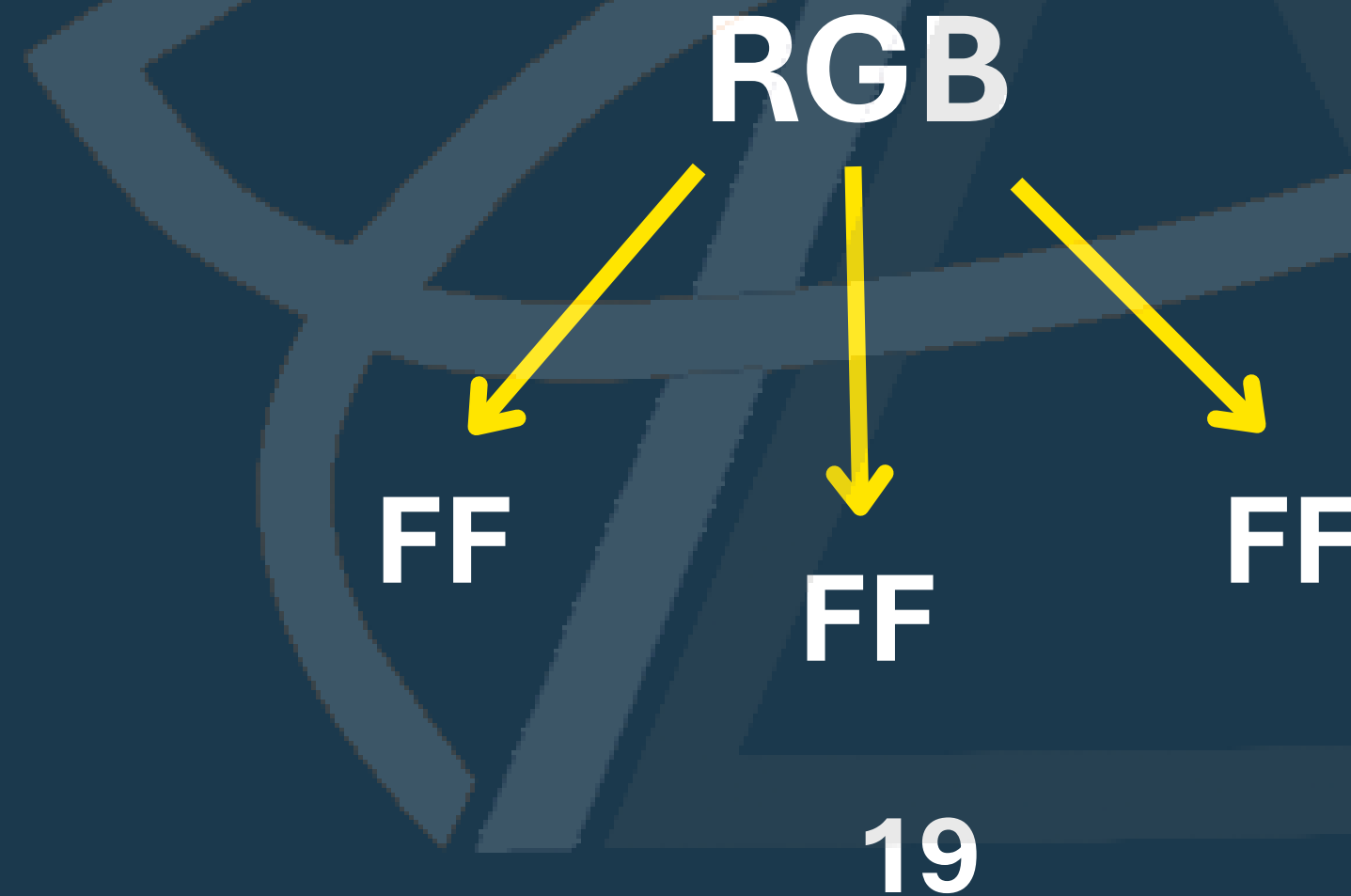
blue is `rgb(0, 0, 255, 0.8)`



Hexadecimal Colors

- Represent colors with a six-digit code, where each pair (00–FF) sets the red, green, and blue intensity.
- Widely used for web design; for example, #ff0000 is red, and #00ff00 is green.

Letters: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F





Common **Text** Properties

1. Font Family
2. Font Size
3. Font Weight
4. Font Style
5. Text Align
6. Text Transform
7. Text Decoration
8. Text Shadow
9. Line Height
10. Letter Spacing



Font Family

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- Defines a prioritized list of fonts to use for text display.
- Helps ensure fallback fonts are shown if the first choice isn't available.

Example:

font-family: arial;

font-family: sans-serif, didot;



Backup Font Family



Font Size

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- Controls how large or small the text appears.
- Affects readability and overall visual hierarchy.

Example:

font-size: 16px;



Font **Weight**

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- Sets the thickness or boldness of text.
- Useful for emphasizing headings or key points.

Example:

font-weight: bold;



Font Style

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- Sets the thickness or boldness of text.
- Useful for emphasizing headings or key points.

Example:

font-style: italic;



Text **Align**

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- Aligns text horizontally (left, right, center, justify).
- Commonly used to control layout and readability.

Example:

text-align: center;



Text Transform

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- Changes the case of text: uppercase, lowercase, capitalize.
- Helps create visual style without changing the actual HTML content.

Example:

text-transform: uppercase;



Text **Decoration**

- Adds decorative lines like underline, overline, or line-through.
- Useful for links, emphasis, or marking removed text.

Example:

text-decoration: underline;



Text **Shadow**

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- Applies shadow effects behind text.
- Adds depth or visual interest to headings and banners.

Example:

text-shadow: 2px 2px 4px rgba(0,0,0,0.5);



Line Height

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- Controls spacing between lines of text.
- Improves readability, especially in paragraphs.

Example:

line-height: 1.5;



Letter Spacing

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- Adjusts the space between individual characters.
- Helps improve legibility or create stylistic effects.

Example:

letter-spacing: 2px;



Let's **recall** what we've learned so far

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